NMOS amplifier Pareto front

Trading gain, bandwidth and supply power with NSGA-II on a Level 3 NMOS stage — milestone M9 (CLAUDE.md §4, §9)

kronspice simulation report · 2026-11-07 · seed 1, NSGA-II 100 × 100

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Summary

This is the M9 acceptance example: the MOSFET levels 1–3 and the NSGA-II multi-objective search used together. A common-source NMOS stage on 5 V, driven from a 1 kΩ source into a 100 kΩ || 5 pF load, is evolved for three objectives at once — gain, bandwidth and supply power — under the constraint that the gain is at least 20 dB. No weights are traded against each other; the run returns the set of designs none of which is beaten on every objective by another (the Pareto front).

- **The front holds 99 designs** after 100 generations (seed 1); in generation 0 only one random design of 100 met the 20 dB constraint.
- **Highest gain:** 34.45 dB at 1.10 MHz bandwidth and 404 µW.
- **Widest band:** 33.8 MHz at 20.17 dB and 19.4 mW.
- **Lowest power:** 22.2 µW at 24.10 dB and 344 kHz.
- **Bandwidth costs power,** and the gain tops out near 35 dB on every seed: the 100 kΩ load caps the drain resistance and gm/I_D cannot exceed its weak-inversion value. Seeds 1–6 reach 34.5–35.2 dB, 34–36 MHz at 17–21 mW, and 20–34 µW.
- **Every front netlist is re-simulated from scratch** by TestMOSParetoExample (constraint, recorded errors, energy ledger, mutual non-dominance).

The circuit

circuit.cir: VDD 5 V; VIN through RSIG 1 kΩ and CIN 1 µF into the gate; the gate bias divider R1/R2; RD to the drain; RSS bypassed by CSS 100 µF at the source; M1 (drain d, gate g, source src, bulk 0, model nproc); COUT 1 µF into RL 100 kΩ || CL 5 pF; AC sweep .ac dec 10 100 1g. The starting values are R1 1 MΩ, R2 500 kΩ, RD 10 kΩ, RSS 1 kΩ and W/L = 20 µm / 2 µm.

The process model is SPICE **Level 3** (semi-empirical) with weak inversion (NFS), so low-power designs biased near or below threshold follow the exponential subthreshold law instead of Level 1's square law. Its parameters are derived from TOX, NSUB and UO as in SPICE3: TOX 20 nm, NSUB 3e16, UO 550, VMAX 1.5e5, THETA 0.06, ETA 0.05, KAPPA 0.4, NFS 4e11, plus overlap and junction capacitances — the same n-channel process as the mos3_* golden circuits. The Level 3 current is written in forward-mode dual numbers, so g_m, g_{ds} and g_{mbs} are the exact derivatives of the stamped current; the gate capacitances are Meyer's.

Genes, objectives and search

gene	range	series
R1, R2 (gate bias divider)	100 kΩ – 10 MΩ	E24
RD (drain load)	100 Ω – 1 MΩ	E24
RSS (bypassed source resistor)	10 Ω – 100 kΩ	E24
M1 W (instance parameter M1.w)	2 – 500 µm	continuous
M1 L (instance parameter M1.l)	1 – 10 µm	continuous

objective	kind	meaning
gain at 10 kHz, V(o)/V(in)	gain_max, ceiling 40 dB	more is better (error = shortfall/20)
upper −3 dB edge	bandwidth_max, ceiling 1 GHz	more is better (error = decades short)

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objective	kind	meaning
power from VDD at the operating point	supply_power, weight 1000 (mW)	less is better
gain at 10 kHz ≥ 20 dB	gain_max with "constraint": true	a constraint, not an objective: a design below 20 dB is dominated by every design above it

GA settings (config.json): algorithm nsga2, population 100, 100 generations, mutation 0.2, σ 0.15, seed 1, timeout 2000 ms per evaluation.

NSGA-II (Deb et al. 2002, evo/nsga2.go):

- **Ranking:** fast non-dominated sort (§III-A) and crowding distance (§III-B); ties are broken by index, so runs are bit-identical across worker counts and resume exactly from a checkpoint.
- **Constrained dominance (§VI):** feasible designs dominate violators, violators are ordered by total weighted violation, and failed simulations (singular matrix, no convergence, timeout) come last, ordered by their M5 penalty class. A failure never crashes the run.
- **Loop (§III-C):** binary crowded tournament, BLX-α crossover and Gaussian mutation as in the weighted GA, then the best population of parents plus offspring by rank and crowding; elitist by construction.
- **Benchmarks** (TestNSGA2ZDT, 10 variables, population 60, 150 generations): on ZDT1 the 60 front members lie within 0.0065 of the true front; on ZDT2 (non-convex, where a weighted sum can only find the two ends) within 0.0040 with 31 interior points.

Results (seed 1)

Five of the 99 front designs:

design	gain	bandwidth	power	RD	RSS	W / L
highest gain	34.45 dB	1.10 MHz	404 µW	62 kΩ	180 Ω	500 µm / 1.84 µm
widest band	20.17 dB	33.8 MHz	19.4 mW	1 kΩ	130 Ω	499 µm / 1.22 µm
lowest power	24.10 dB	344 kHz	22.2 µW	1 MΩ	27 kΩ	301 µm / 6.47 µm
≥ 30 dB at least power	31.41 dB	392 kHz	56.7 µW	430 kΩ	910 Ω	427 µm / 3.43 µm
≥ 10 MHz at least power	23.49 dB	11.7 MHz	2.88 mW	3.3 kΩ	750 Ω	500 µm / 1.32 µm

- **Bandwidth costs power:** a small RD against the 5 pF load needs current to keep the gain.
- **The gain ceiling** is near 35 dB because the gain is $g_m \cdot (RD \parallel RL \parallel r_o)$: the 100 kΩ load caps the drain resistance, and g_m/I_D is at most its weak-inversion value $1/(n \cdot V_T)$. The highest-gain designs use the widest device the genes allow, in moderate inversion.
- **Across seeds 1–6:** highest gain 34.5–35.2 dB, widest band 34–36 MHz at 17–21 mW, lowest power 20–34 µW, all at gain ≥ 20 dB.

Besides the usual GA outputs the run writes front.csv (every front design with its three errors, the constraint and the gene values, sorted by the first objective) and front/NNN.cir, one netlist per design, each of which opens in ngspice, LTspice and QSpice as it is. No front plot is stored in the example folder.

Validation

- **TestMOSParetoExample** (evo/mos_pareto_test.go) re-simulates every front netlist from scratch and checks the constraint, the recorded errors, the energy ledger and mutual non-dominance, and that the front spans the trade-off (a design above 34 dB, one above 30 MHz, one below 50 µW).
- **TestNSGA2ZDT** checks the search itself on the ZDT1 and ZDT2 benchmark fronts (numbers above).
- The export/ folder holds the starting circuit as .cir, .kicad_sch and a results sheet (MNA only: Kron's torn solve on that sheet is linear-only, and M1 is nonlinear).

Limits

- One generic 1 µm-class Level 3 process; no process corners or mismatch.
- Gain is scored at one frequency (10 kHz) and power at the DC operating point only.
- Small-signal AC analysis: no distortion or output swing objective.

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Reproducing the results

```
go run ./cmd/evolve -config examples/mos_amplifier_pareto/config.json   # about 6 s
go test ./evo -run 'TestMOSParetoExample|TestNSGA2'                       # acceptance and ZDT benchmarks
```


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